

HT37X – Extended Reconfigurable Plate Heat Exchanger

The HT37X is designed to be reconfigurable by the student and can accommodate up to four sections of heating.

PRODUCT CODE: HT37X

Categories: Educational, HT Series - Computer Controlled Heat Exchange / Heat Transfer, HT30X Service Unit Heat Exchange Accessories

Tags: 316 Stainless Steel , Armbus , Extended Reconfigurable Plate Heat Exchanger , Heat Exchanger , HT37 , HT37X , Parallel Passes , Regeneration , Serial Passes , Thermocouples

DESCRIPTION

The HT37X is designed to be reconfigurable by the student and can accommodate up to four sections of heating. Each section provides an additional temperature measurement point for each fluid stream.

In order to make the unit easy to reconfigure, these sections are supplied as pre-assembled groups of plates complete with an intermediate plate (containing the temperature measurement points). Using the four heating sections provided, students can compare heat exchangers of different heat transfer area and different numbers of passes. A quick release clamp system enables the different arrangements to be changed quickly and easily without using tools.

Also supplied is a pre-assembled regeneration section to demonstrate this important energy efficient method of heating. The regeneration section can be used in conjunction with one, two or three of the standard heating sections in different configurations.

Plate heat exchangers can be implemented in a wide variety of configurations, with parallel passes, serial passes, or combinations of both. For more advanced investigations into these effects, (e.g. for project work) a further twelve loose plates are supplied with the HT37X.

These can be used in conjunction with the plates from the pre-assembled modules to investigate additional configurations.

TECHNICAL SPECIFICATIONS

Number of plates:

- 5, 10, 15 or 20 (no regeneration)
- 6, 11, 16 or 21 (with regeneration)

Heat transfer area: 0.096m²

Heat exchanger configuration: Configurable sections, with a separate regeneration section

Thermocouples from Accessory:

- Hot fluid interim positions x3
- Cold fluid interim positions x3

Thermocouples on Service Unit:

- Hot fluid inlet
- Hot fluid outlet
- Cold fluid inlet
- Cold fluid outlet

Clamping mechanism: Dual screw threads with nuts

FEATURES & BENEFITS

- A small scale plate heat exchanger system for use with an Armfield Computer Controlled Heat Exchange Service Unit to teach the fundamental concepts of heat exchangers
- Comprises a number of stainless steel plates, each with a pressed chevron pattern and a food grade silicon rubber sealing gasket, mounted in a frame
- Easily dismantled for cleaning
- A comprehensive instruction manual is included

Technical Details

The plates used in these heat exchangers have been specifically developed by Armfield for use in miniature heat exchanger systems. They are fabricated from 316 stainless steel, with a pressed chevron pattern to promote turbulence and provide multiple support points.

Silicone rubber gaskets are used on each plate to seal the adjacent flow channels from each other.

Plate details:

- Plate overall dimensions: 75mm x 115mm
- Effective diameter: 3.0mm
- Plate thickness: 0.5mm
- Wetted perimeter: 53.0mm
- Projected heat transmission area: 0.008m² per plate

The plates are mounted in a frame incorporating fixed and moving end plates with connections for hot and cold fluids. The exchangers are easily dismantled for inspection of the heat transfer surfaces.

armBUS Software Features

ArmBUS software is provided as standard with the HT30X for all the Armfield heat exchanger accessories. Individual programs are provided for each exchanger, and each program contains a selection of separate exercises that can be performed.

The actual details are exercise specific, but typically the following interfaces are available:

- Full Graphical User Interface (GUI) for each experiment with display screens tailored for each exercise
- All the temperatures and flow rates are displayed on a diagrammatic representation of the equipment
- Student questions and equations can be accessed individually for each exercise



- The hot-water flow rate utilises optional Manual/Automatic control to control the pump output. Manual control allows the user to select a percentage of the total power to supply to the pump. Automatic control allows the user to input the desired set point which the PID control will reach and maintain. The use of automatic control ensures the flow is stable despite changes in the viscosity of the water due to heating
- Real-time flow rate readings are displayed in L/min Sensor data is collated and calculations are displayed in a data-log, a tabulation function provided with the armBUS software. The data is in tabulated format and can be saved and accessed through a .csv file compatible with software such as Microsoft Excel
- The cold-water flow control valve utilises optional Manual/Automatic settings to control water flow from the mains. Manual control allows the user to open/close the valve between 0 and 100%. Automatic control allows the user to input the desired set point which the PID control will reach and maintain so long as the input pressure is adequate.
- The heater utilises optional Manual/Automatic control to supply power to the heater element in the boiler. Manual control allows the user to select a percentage of the total power to supply to the heater. Automatic control allows the user to select a set point (°C), and then PID control software heats the water to and maintains a set temperature
- The data from the sensors is plotted and displayed in a user-configurable graphing function of the software. It can be displayed separately from the data logging and each sensor output can be viewed independently
- Processing of measured values to obtain calculated results (this can be linked to the questions and answers to ensure student understanding, answers to equation calculations can be found as part of the datalogger output)

Essential Accessories

HT30X Computer Controlled Heat Exchanger Service Unit

Shipping Specifications

PACKED AND CRATED SHIPPING SPECIFICATIONS:

Volume: 0.03m3

Gross weight: 6 kg

Dimensions

Length: 0.17m

Width: 0.18m

Height: 0.39m

Order Codes

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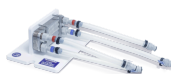
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